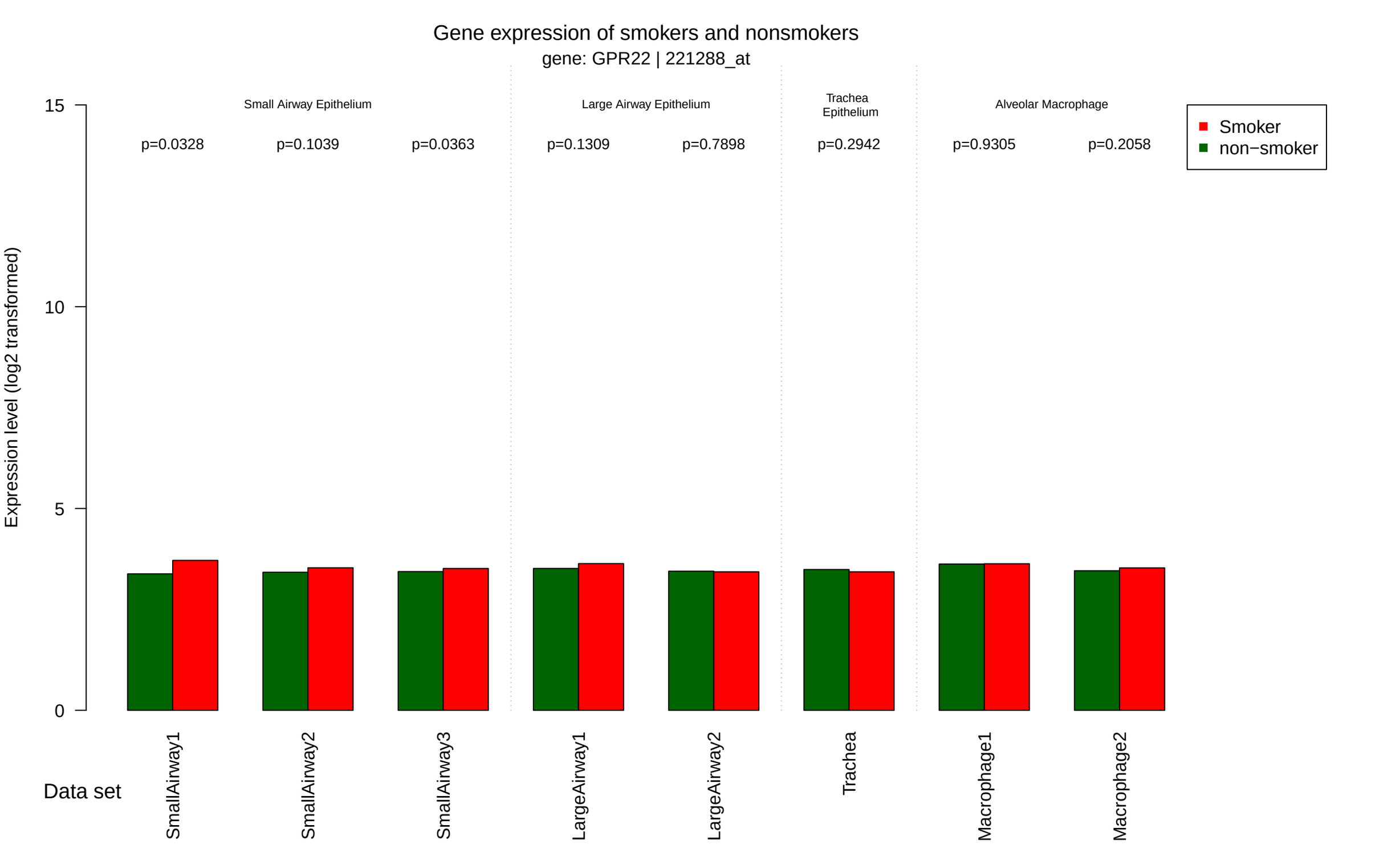
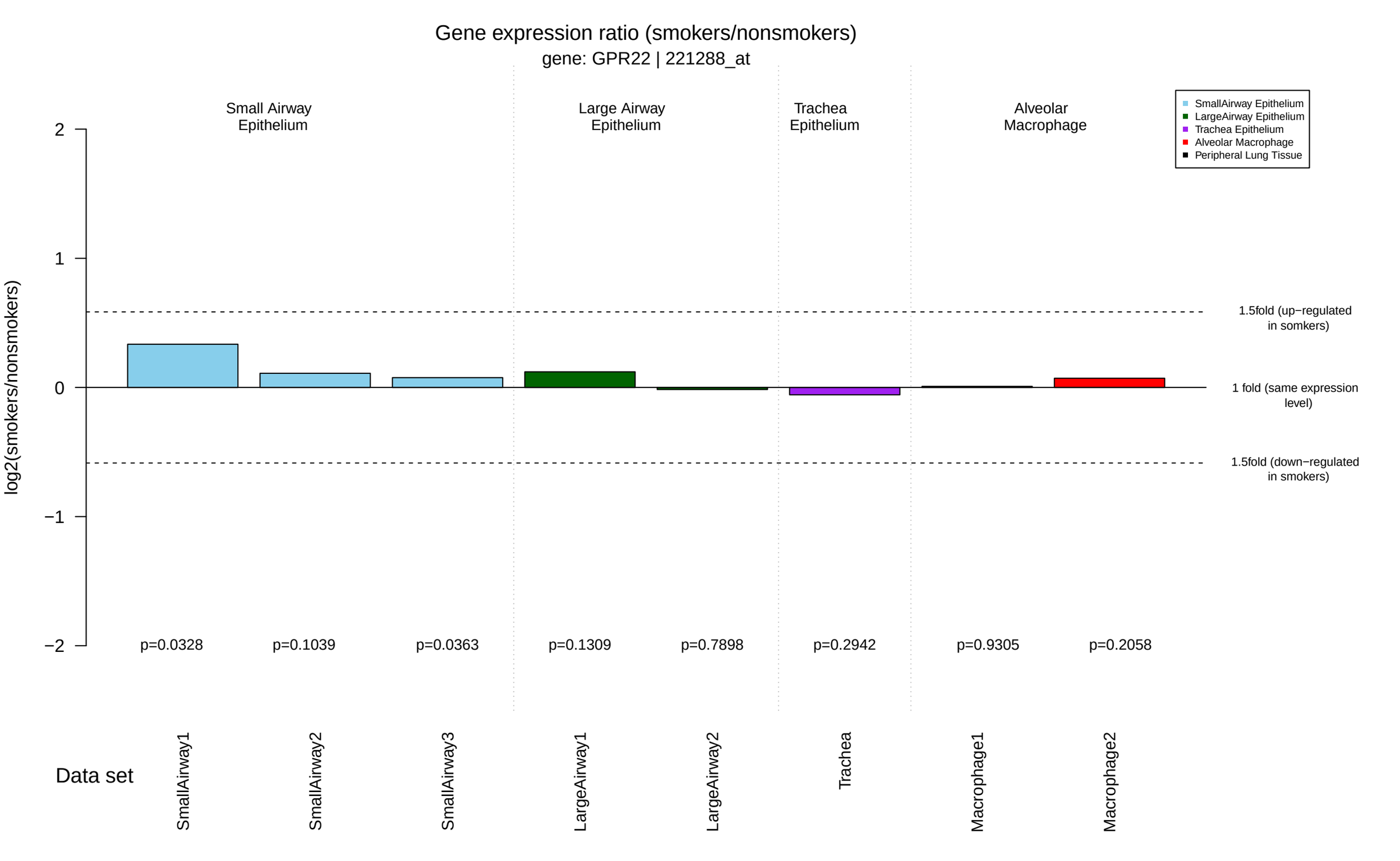
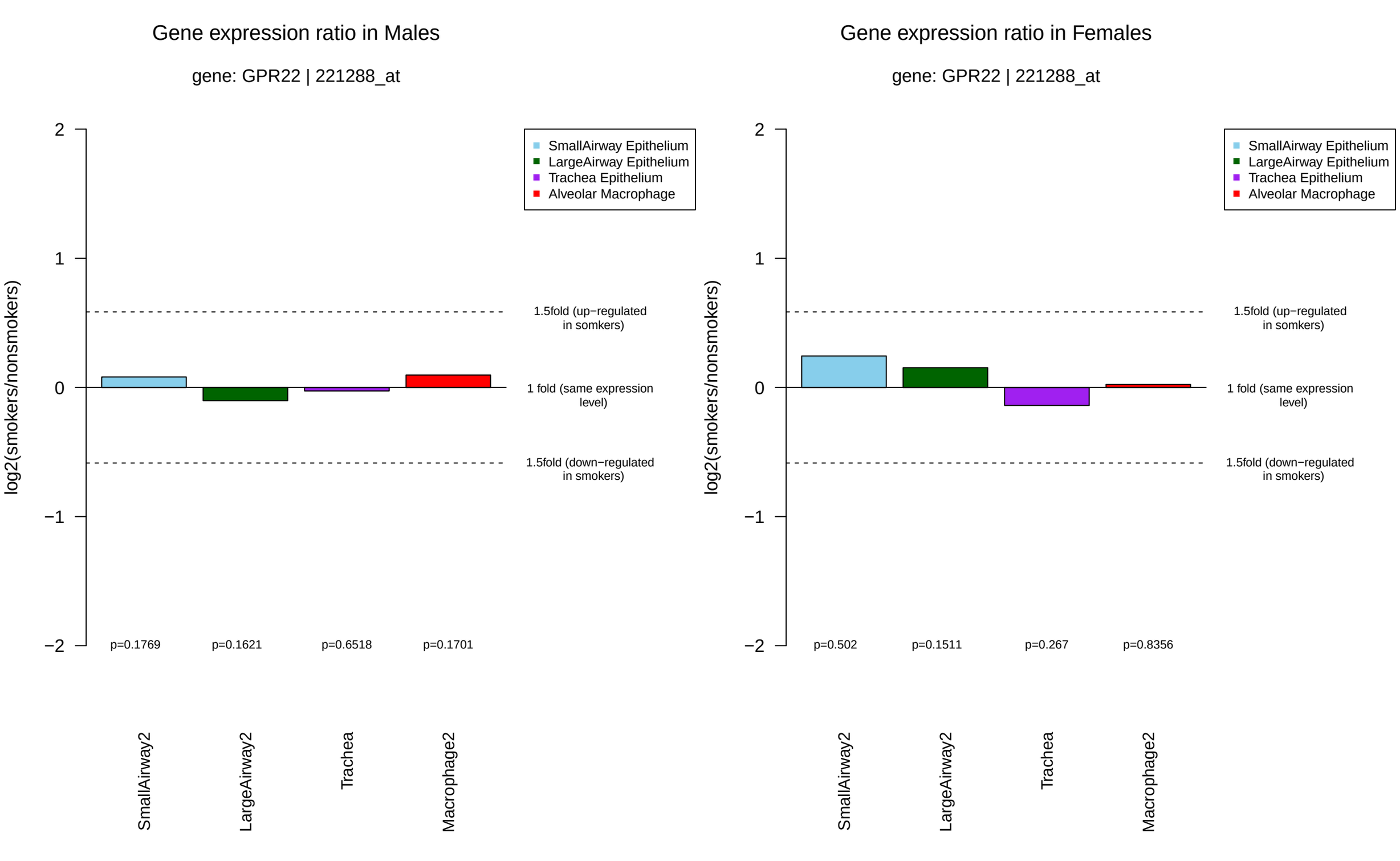
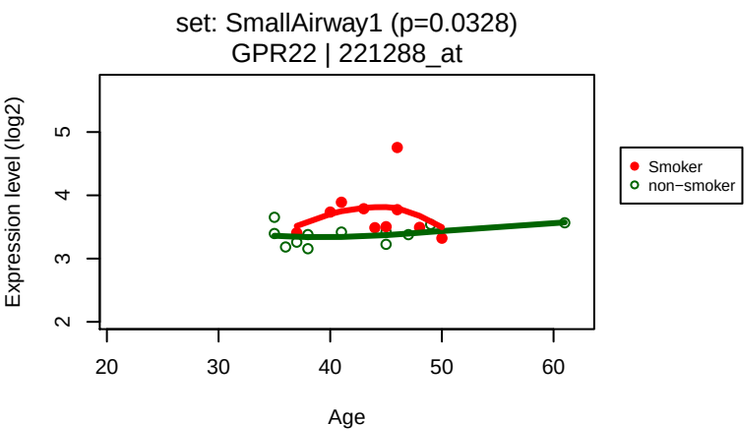
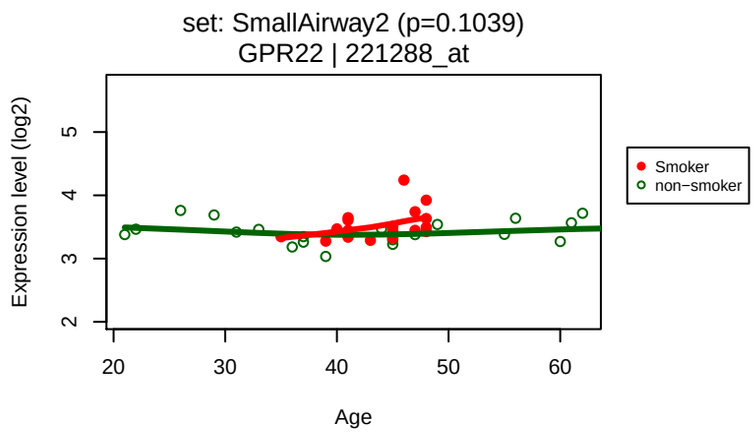
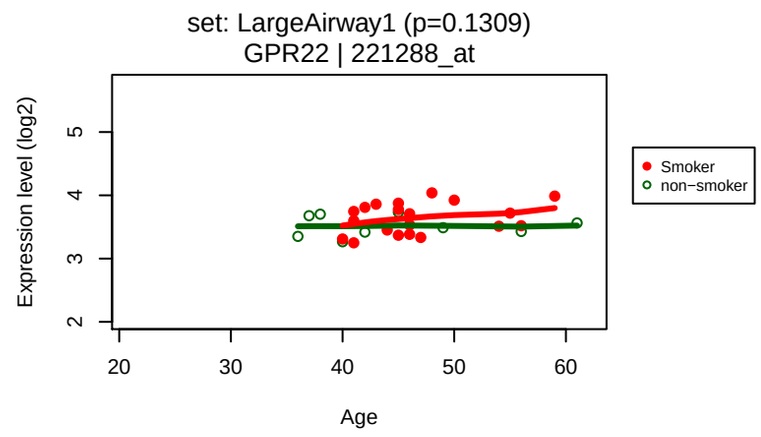

Gene expression of smokers and nonsmokers
gene: GPR22 | 221288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0328
p=0.1039
p=0.0363
p=0.1309
p=0.7898
p=0.2942
p=0.9305
p=0.2058
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

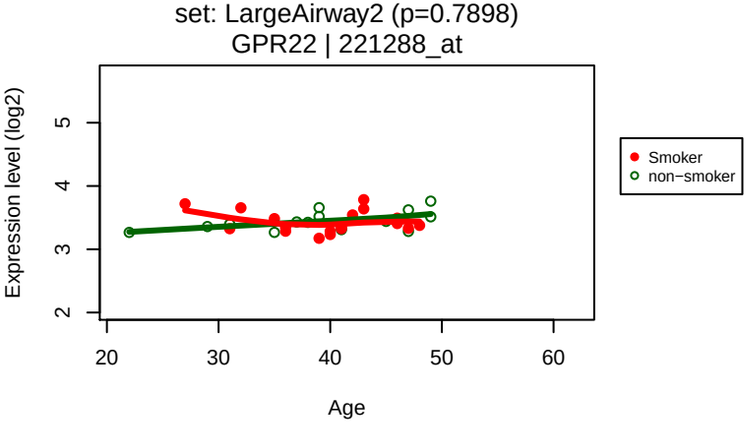

Gene expression ratio (smokers/nonsmokers)
gene: GPR22 | 221288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0328
p=0.1039
p=0.0363
p=0.1309
p=0.7898
p=0.2942
p=0.9305
p=0.2058
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

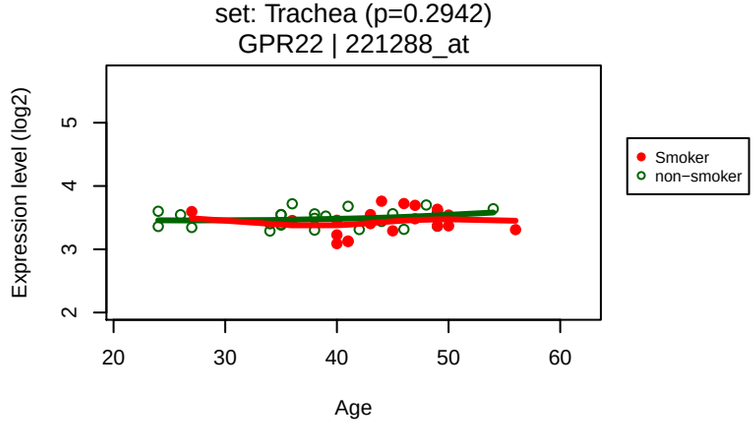

Gene expression ratio in Males
gene: GPR22 | 221288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1769
p=0.1621
p=0.6518
p=0.1701
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR22 | 221288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.502
p=0.1511
p=0.267
p=0.8356
SmallAirway2
LargeAirway2
Trachea
Macrophage2

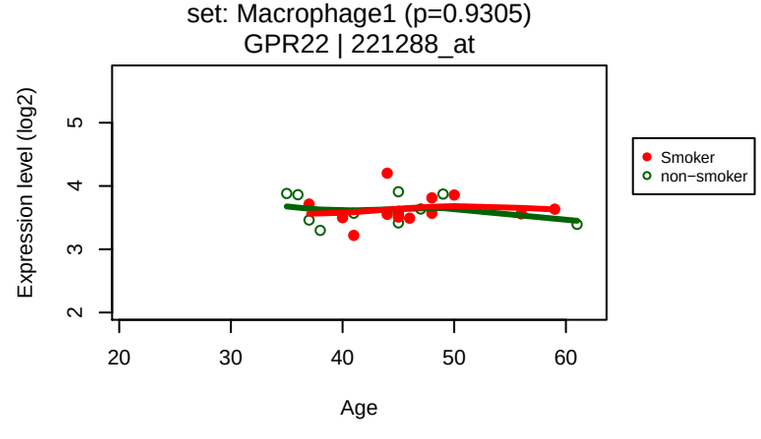

set: SmallAirway1 (p=0.0328)
GPR22 | 221288_at
Expression level (log2)
Age
Smoker
non−smoker

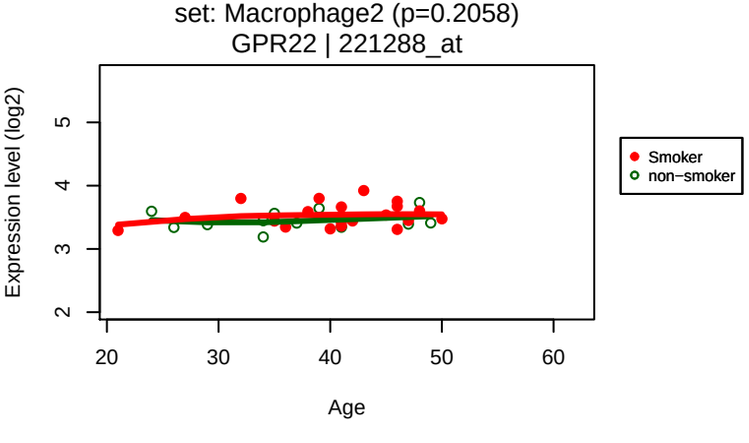

set: SmallAirway2 (p=0.1039)
GPR22 | 221288_at
Expression level (log2)
Age
Smoker
non−smoker

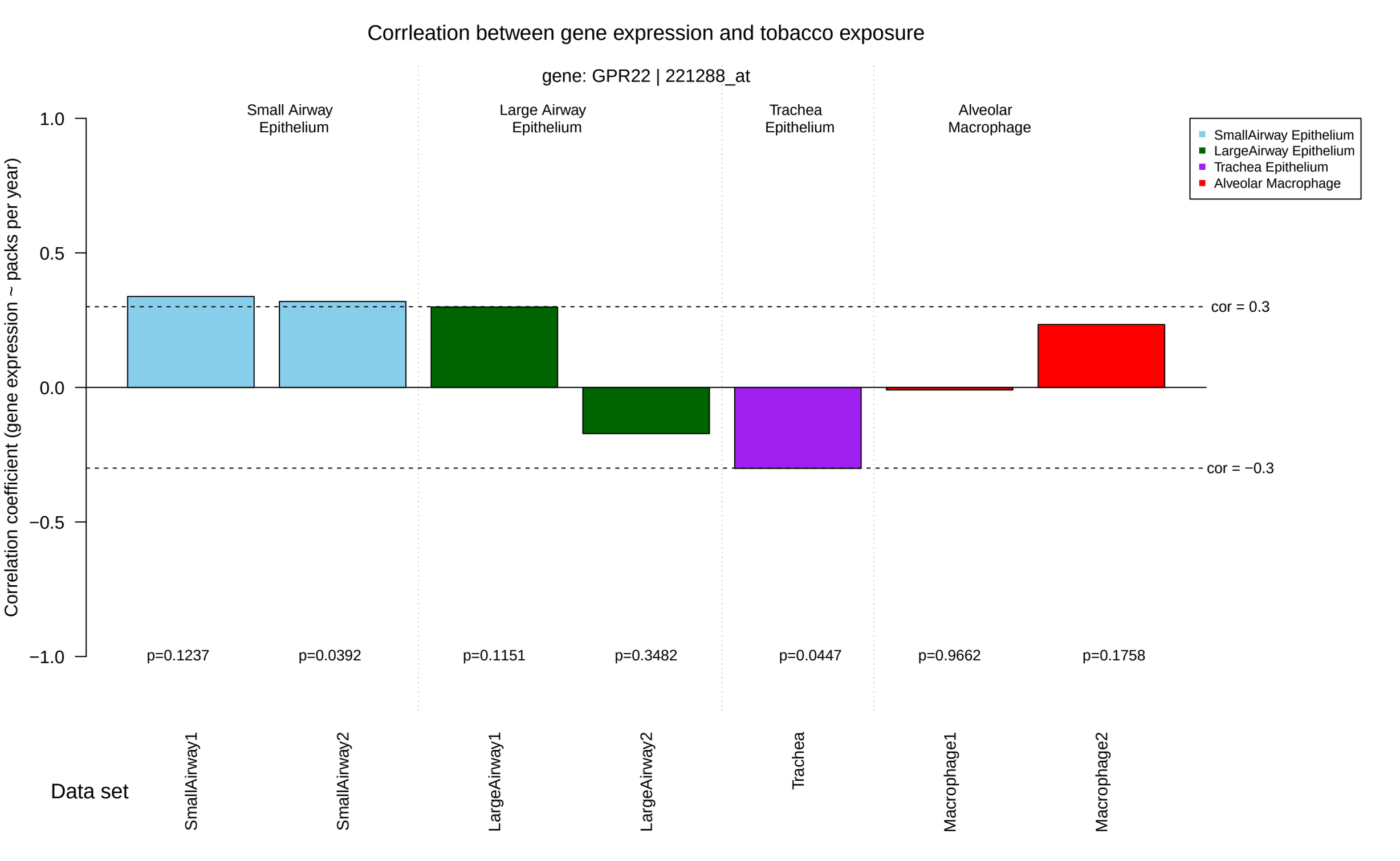

set: LargeAirway1 (p=0.1309)
GPR22 | 221288_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7898)
GPR22 | 221288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2942)
GPR22 | 221288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9305)
GPR22 | 221288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2058)
GPR22 | 221288_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR22 | 221288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1237
p=0.0392
p=0.1151
p=0.3482
p=0.0447
p=0.9662
p=0.1758
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2